تحدي الذكاء الاصطناعي
في خدمة المحتوى الإسلامي

AI CHALLENGE
Serving Islamic Content

باذل
BATHEL



PrayAlong

A 3D companion that prays with you, step by step, and follows your recitation on your own device

October 2026 | prayalong.me

Quran Lab | Track 03: Interactive experiences and learning journey

Learning to pray is hard without a teacher

Salah is the first act of worship a new Muslim needs every day, and it is learned by doing, not by reading.

New Muslims and reverts

Must learn the movements, the Arabic and their order at once, often with no Muslim nearby to copy.

Children

Learn by praying next to a parent, and families cannot be there for every prayer.

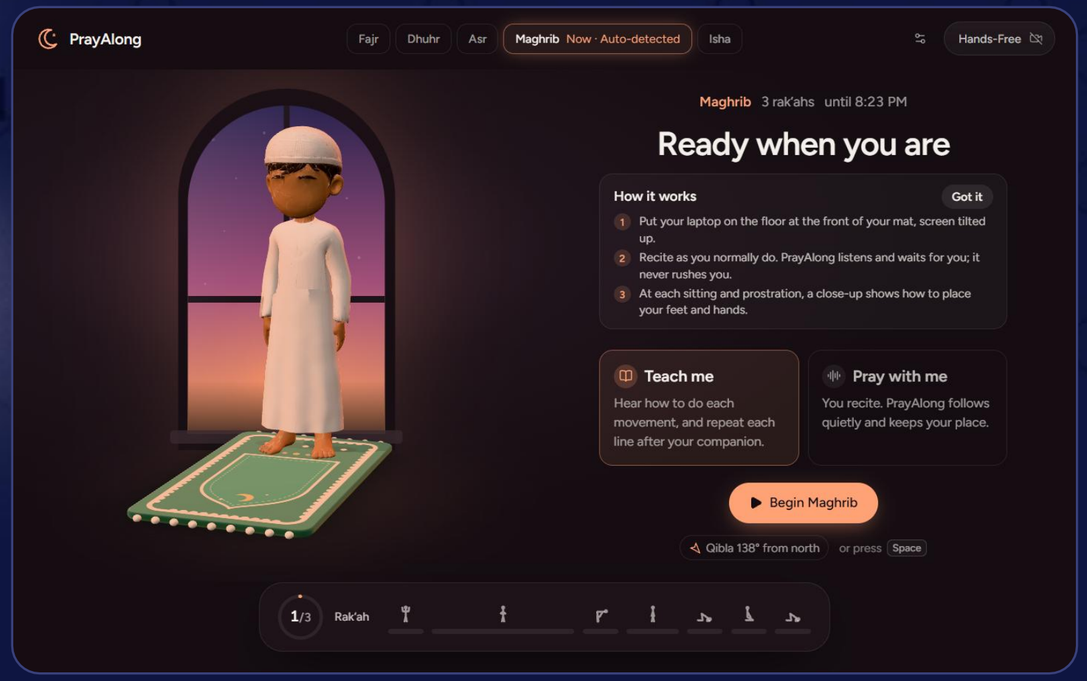
A teacher is not always there

Classes are few, far away, or not in the learner's language. Asking feels awkward for adults.

Apps only play or show

Videos and step lists cannot hear you. They cannot wait for you, keep your place or count with you.

PrayAlong: a companion that prays with you



What you see

A 3D companion on a prayer mat moves through every posture, with the Arabic, transliteration and meaning of each line beside it.

Two modes

Teach me: it explains each movement and recites each line; you repeat.
Pray with me: you recite; it follows quietly and keeps your place.

Try it now at prayalong.me



Scan, open, press Begin

Works in any modern browser on a laptop or a phone. No account, no install.
Allow the microphone and it follows you; the audio never leaves the device.

Code and run docs: github.com/Quran-Lab/PrayAlong

Everything runs on the device

Recited audio (ElevenLabs) is generated once and shipped as files. The optional camera (beta) runs MediaPipe and DETRPose in the browser.

01

Microphone

AudioWorklet, resampled to 16 kHz

Off until you allow it

02

Quran ASR

Zipformer2 CTC phoneme model v3.1 in WebAssembly

160 ms streaming

03

Aligner + conductor

Prayer-graph aligner: where you are, word by word

Moves the prayer forward only

04

3D companion

Highlights each word, counts tasbih, changes posture on the takbir

AI listens and speaks; it never writes the text

No generative model produces Quran, supplications or rulings. Every word shown or recited comes from a named source.

Quran speech recognition

Quran Lab zipformer2 CTC phoneme model (v3.1, int8, 68 MB) in WASM. Phonemes, so it hears learners and tolerates mistakes.

Prayer-graph aligner

Matches what it hears only against the expected next lines: word highlight, tasbih count, takbir detection. It cannot wander off.

Neural voices

ElevenLabs eleven_v4, four voices, rendered once from the source text. Spoken guidance in 9 languages, played offline.

Pose estimation (beta)

MediaPipe Pose and Face plus DETRPose in Web Workers decide when you have moved to the next posture.

Sourced, levelled, and never a fatwa

Each content type is mapped to the organiser’s levels A to D. Details and review status: docs/SOURCES.md in the repo.

Content	Source	Level	How it is verified
Quran text and meanings	Tanzil text; published translations (e.g. Saheeh International, Bubenheim, Diyanet) fetched verbatim	A	Script fetch, never hand-edited; credit shown in app; tests check every line
Prayer supplications	Hisn al-Muslim, with its hadith references (al-Bukhari, Muslim, the Sunan)	A	Hadith to be checked on dorar.net; translations queued for native review
Movements and order	The standard fard form; school differences noted	A / C	Differences are settings with a note, e.g. raising the hands at ruku
Personal rulings	Not provided	D	The app gives no fatwa and refers the user to a scholar

Privacy: no account, no tracking, no upload. Audio and video are processed in the browser and discarded. The app infers nothing about the user’s faith.

Measured on the real app, end to end

Recorded prayers played into the live app through a fake microphone, in real time (Chromium, all layers on).

8/8

Prayers completed

Standard set, 0 early moves, incl. noise at 5 dB SNR

72/72

Tasbih counted right

Plus 6/6 perfect on the amin set

78 → 7

Early moves

Real-like set (tempo, pitch, room, quiet), after the fixes

4.9%

Takbir phoneme error

Down from 9.4% with in-worker 16 kHz resampling

Also: word lag 0.38 s (p50); camera beta: 70.8% of transitions, 0 false advances. Sources: docs/voice.md, docs/hands-free.md

Nine languages, right to left, phone and laptop

English, Arabic, Urdu, Turkish, Indonesian, French, German, Spanish, Dutch. Two modes, reduced motion, no sign-up.



Full Arabic interface, right to left





Phones: prop it on the floor in front of you



Close-ups for the feet and the tashahhud finger

A prayer companion that listens and follows you

Not a video to watch: a companion that waits for you, keeps your place and moves when you move.



It listens

On-device Quran ASR follows recitation word by word, counts tasbih, hears the takbir.



It teaches the body

3D postures from one rig, with close-ups for the feet and the finger.



Halal by design

Four modest companions with eyebrows-only faces: man, woman, boy, girl.



Private by design

Model runs in the browser. No server sees a voice or a face.

From hackathon to everyday use

Next 12 months. Free, with all source code public (NPL-1.2), low running cost: static site plus model files on a CDN, no inference servers.

01

Scholar review

Months 1-2

Arabic supplications and all translations signed off; quranenc.com editions

02

Real-user pilot

Months 2-4

Mosques and revert centres; measure learning and completion

03

Camera by default

Months 4-6

From beta to on by default after real-recording tests

04

More prayers

Months 6-9

Sunnah, witr and Jumu’ah guidance

05

Offline PWA

Months 9-12

Install once, pray anywhere without a connection

Quran Lab

Quran Lab builds Quranic speech technology with public source code. PrayAlong reuses that stack to teach the prayer.

Mostafa Mahdi

Team lead AI/ML engineer, speech and on-device inference

Raufa Zuhdi Aristyo

Team member AI/ML engineer, computer vision and evaluation

Public and verifiable

Code: github.com/Quran-Lab/PrayAlong
Live: prayalong.me

The four companions in the app



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عام الذكاء الاصطناعي
THE YEAR OF ARTIFICIAL INTELLIGENCE 2026

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Thank you

PrayAlong by Quran Lab

prayalong.me | github.com/Quran-Lab/PrayAlong

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